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328813 (28)

BE (8th Semester)

Examination, April-May, 2013

Branch : Et & T

INDUSTRIAL & POWER ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory and carry 2 marks.

(ii) Attempt any two from parts (b), (c) and (d) each carry 7 marks.

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Q. 1. (a) Give the basic principle of operation of SMPS ? 2

(b) A zener diode is rated at 50 V over a range of diode current from 5 to 40 mA. The supply voltage is 200 V. Calculate the resistance R_s to be connected in series with the power supply and zener diode to allow voltage regulation from a load. $I_L = 0$ to $I_{L\ max}$. Also find the value of maximum load current $I_{L\ max}$. 7

(c) Explain with a neat circuit diagram the operation of buck-boost regulator. 7

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(d) Explain with the help of neat circuit diagram the operation of shunt voltage regulator. 7

Q. 2. (a) Define Holding current and Latching current. 2

(b) Briefly explain two transistor model of SCR and describe the condition for its conduction. 7

(c) For an SCR gate-cathode characteristic is given by $V_g = 1 + 10\ I_g$. Gate source voltage is a rectangular pulse of 15 V with 20 μ sec duration. For an average gate power dissipation of 0.3 W & peak gate power dissipation of 5 W. Calculate : 7

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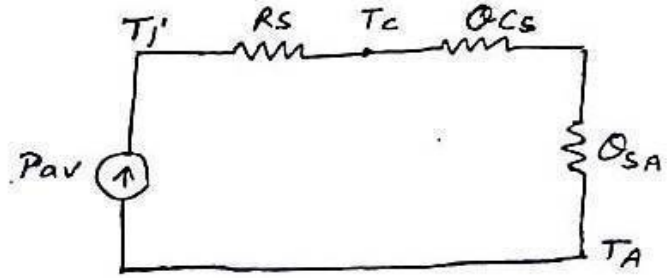
(i) Series resistance of gate

(ii) Triggering frequency

(d) An SCR has continues average rating $I_{t\,av}$ of 25 A and a dynamic resistance R_T of 1 Ω. If the casing temperature is decreased from 40°C to 30°C by efficient cooling. Calculate the percentage increase in the device rating. Assume permissible junction temperature is 125°C. 7



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Q. 3. (a) How are inverters classified ? 2

(b) Describe with the help of circuit diagram circulating current type dual converter. 7

(c) Describe the basic principle of working of single phase to single phase stepdown cycloconverter for continuous conduction of a bridge type cycloconverter. Mark the conduction of various thyristors. 7

(d) Briefly explain the principle of operation of dc choppers and mention some application of chopper in industry. 7

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Q. 4. (a) Define skin effect in Induction heating. Give the cause of it. 2

(b) For the high frequency induction heating derive the total power entering the metal per sq cm of the surface given by 7

$$P_t = \frac{8\pi H_o \sqrt{10^{-9} \mu_r . f}}{\sqrt{\sigma}}$$

(c) Explain different types of resistance welding. 7

(d) A metal cylinder 5 cm long and 1 cm diameter is placed inside a coil having 10 turns. The coil carries current of 100 A. If the depth of heating is 0.4 mm, determine the

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frequency and also heat generated per unit surface area. Assume $\mu_r = 1$ and resistivity of metal to be 5×10^{-8} Ω-m. 7

Q. 5. (a) Explain the working principle of servomotor. 2

(b) Explain different types of noise. 7

(c) Briefly explain with the help of simplified block diagram operation of ON line and OFF line u.p.s. 7

(d) Briefly explain with the help of block diagram the working principle of servo controlled voltage stabilizer. 7

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